MIS 443 – BUSINESS DATA MANAGEMENT

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**Business Scenario and Dataset**

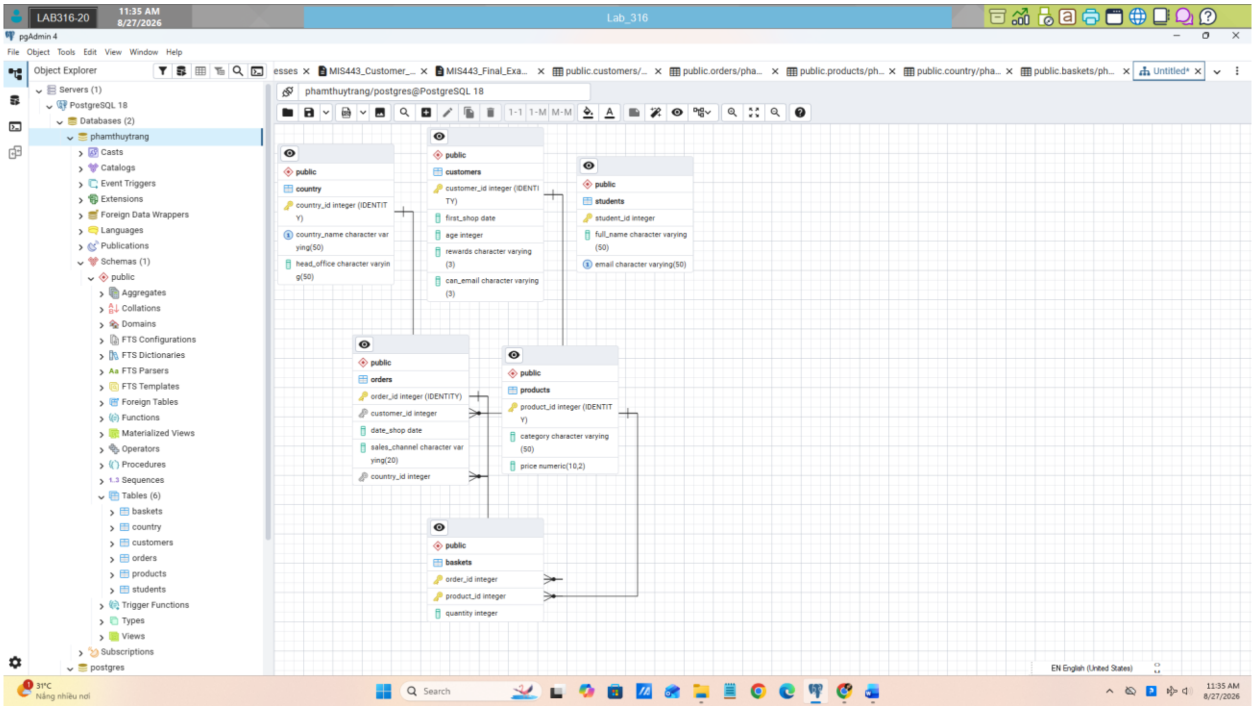
The Customer Insights database represents an international retailer selling products through retail and online channels. Management uses the database to analyse customers, orders, markets, product categories, and sales performance.

***The database contains five tables:***

- customers: customer age, first purchase, rewards membership, and email permission;
- orders: order date, customer, sales channel, and country;
- country: country name and head-office location;
- products: product category and unit price;
- baskets: products and quantities included in each order.

***Key relationships:***

- One customer can place many orders.
- One country can have many orders.
- Orders and products have a many-to-many relationship through baskets.

- 

**Question 1 – Database Setup (10 marks)**

***(a) Load the database – 5 marks***

Create a PostgreSQL database using your full name in lowercase, without spaces or Vietnamese diacritics. Connect to the database and execute:

MIS443_Customer_Insights_PostgreSQL.sql

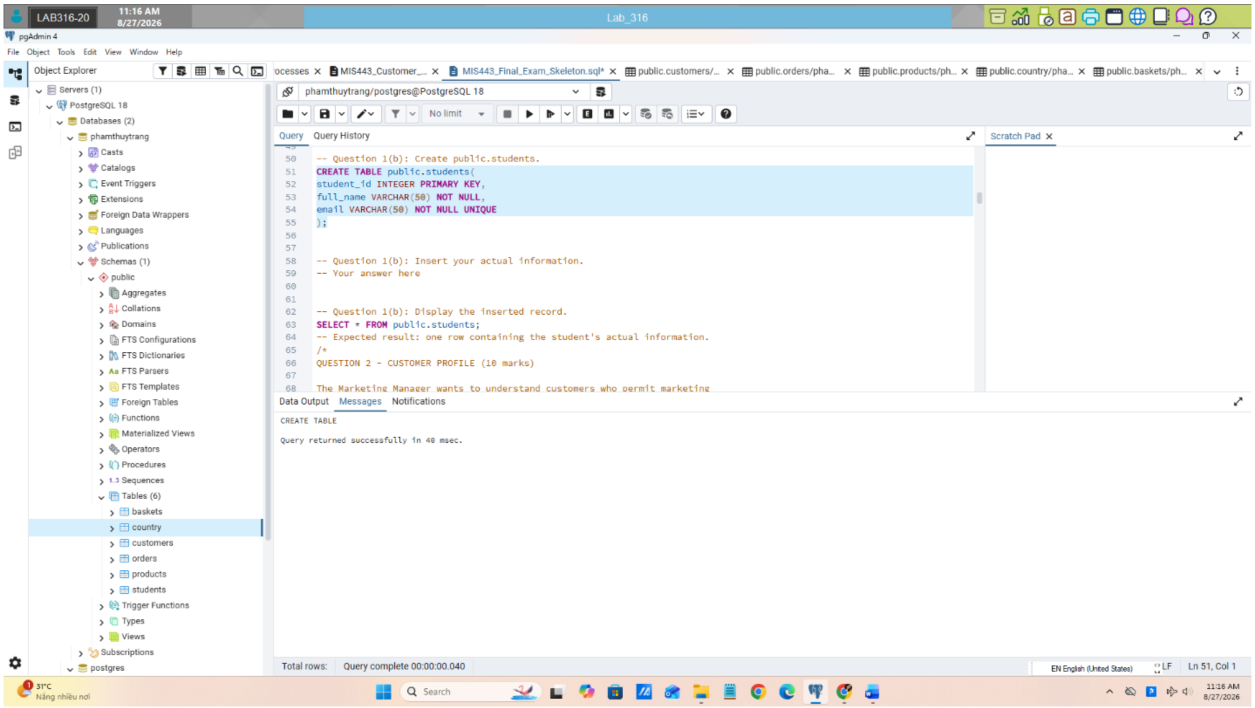
Confirm that country, customers, orders, products, and baskets are available in the public schema.

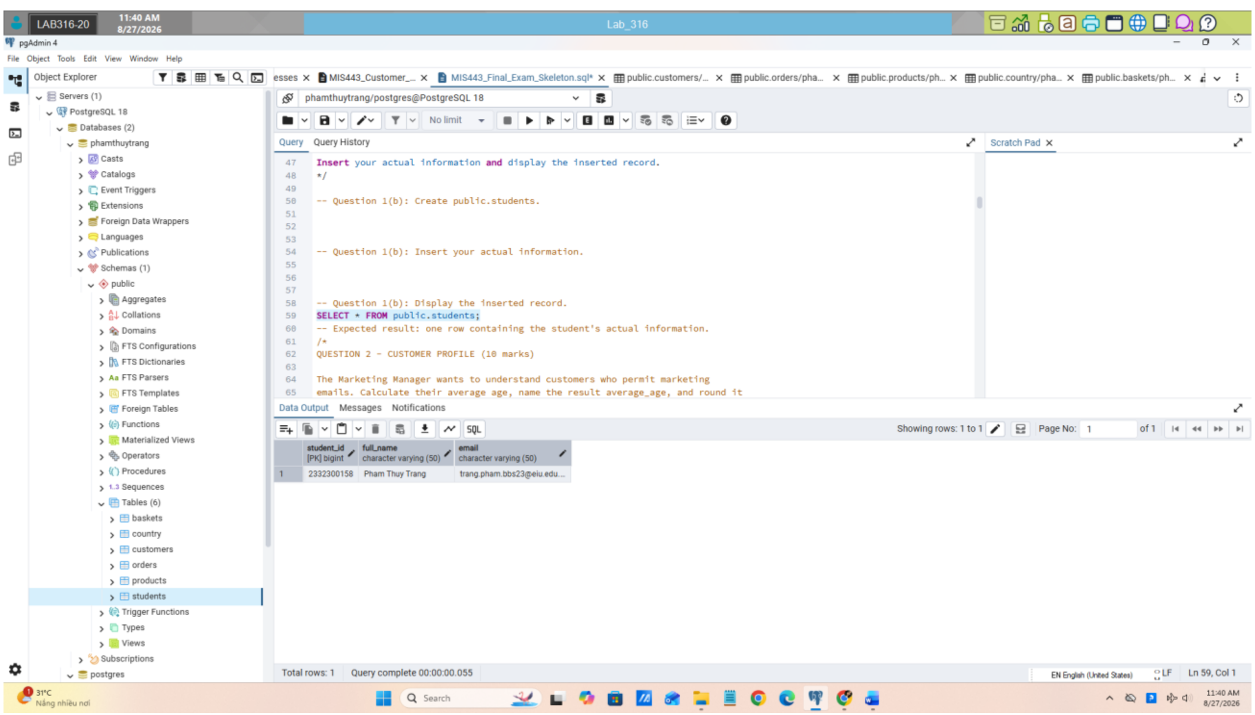
***(b) Create the student table – 5 marks***

Create public.students with:

- student_id: exactly 10 characters and primary key;
- full_name: required;
- email: required and unique.

Insert your actual information and display the inserted record.


LAB316-20
11:16 AM
8/27/2026
Lab_316
pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
MIS443_Final_Exam_Skeleton.sql*
phamthuytrang/postgres@PostgreSQL 18
Query Query History
-- Question 1(b): Create public.students.
CREATE TABLE public.students(
student_id INTEGER PRIMARY KEY,
full_name VARCHAR(50) NOT NULL,
email VARCHAR(50) NOT NULL UNIQUE
);
-- Question 1(b): Insert your actual information.
-- Your answer here
-- Question 1(b): Display the inserted record.
SELECT * FROM public.students;
-- Expected result: one row containing the student's actual information.
/*
QUESTION 2 - CUSTOMER PROFILE (10 marks)
The Marketing Manager wants to understand customers who permit marketing
Data Output Messages Notifications
CREATE TABLE
Query returned successfully in 40 msec.
Total rows: Query complete 00:00:00.040
Ln 51, Col 1


LAB316-20
11:40 AM
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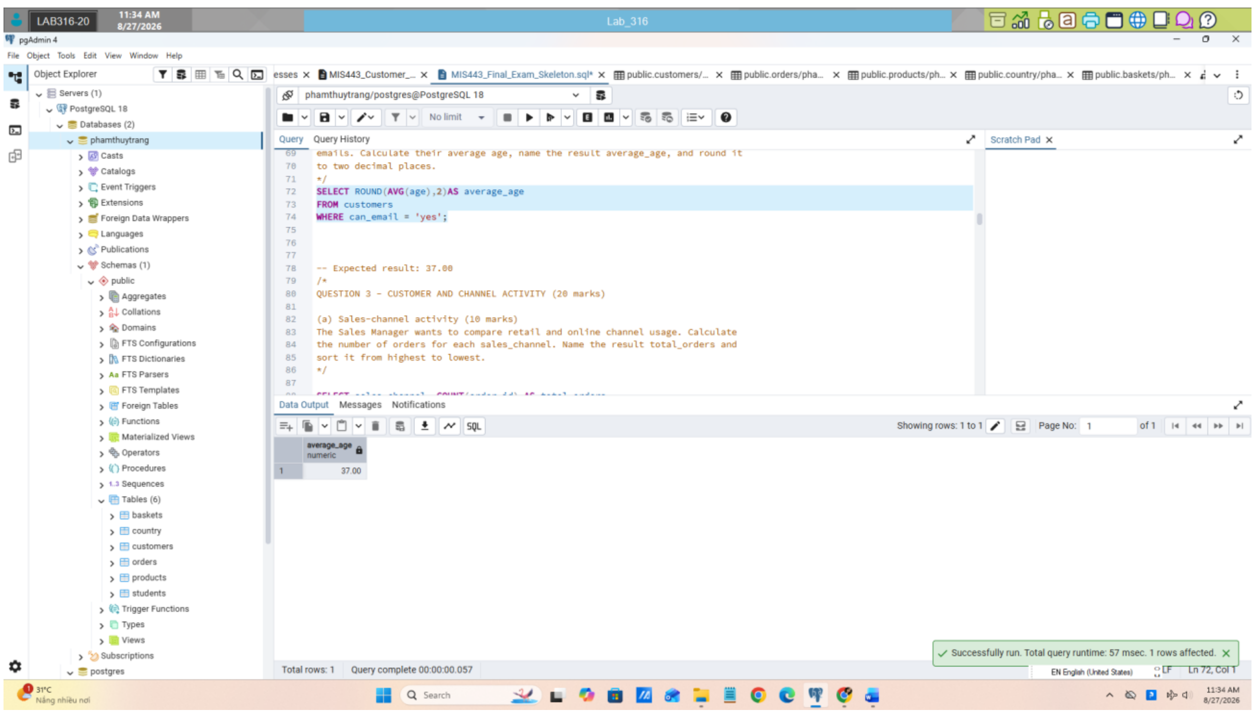
Lab_316
pgAdmin 4
MIS443_Final_Exam_Skeleton.sql*
phamthuytrang/postgres@PostgreSQL 18
Insert your actual information and display the inserted record.
*/
-- Question 1(b): Create public.students.
-- Question 1(b): Insert your actual information.
-- Question 1(b): Display the inserted record.
SELECT * FROM public.students;
-- Expected result: one row containing the student's actual information.
/*
QUESTION 2 - CUSTOMER PROFILE (10 marks)
The Marketing Manager wants to understand customers who permit marketing
emails. Calculate their average age, name the result average_age, and round it
Data Output Messages Notifications
Showing rows: 1 to 1
student_id [PK] bigint
full_name character varying (50)
email character varying (50)
1 2332300158 Pham Thuy Trang trang.pham.bbs23@eiu.edu...
Total rows: 1 Query complete 00:00:00.055
Ln 59, Col 1

### Question 2 – Customer Profile (10 marks)

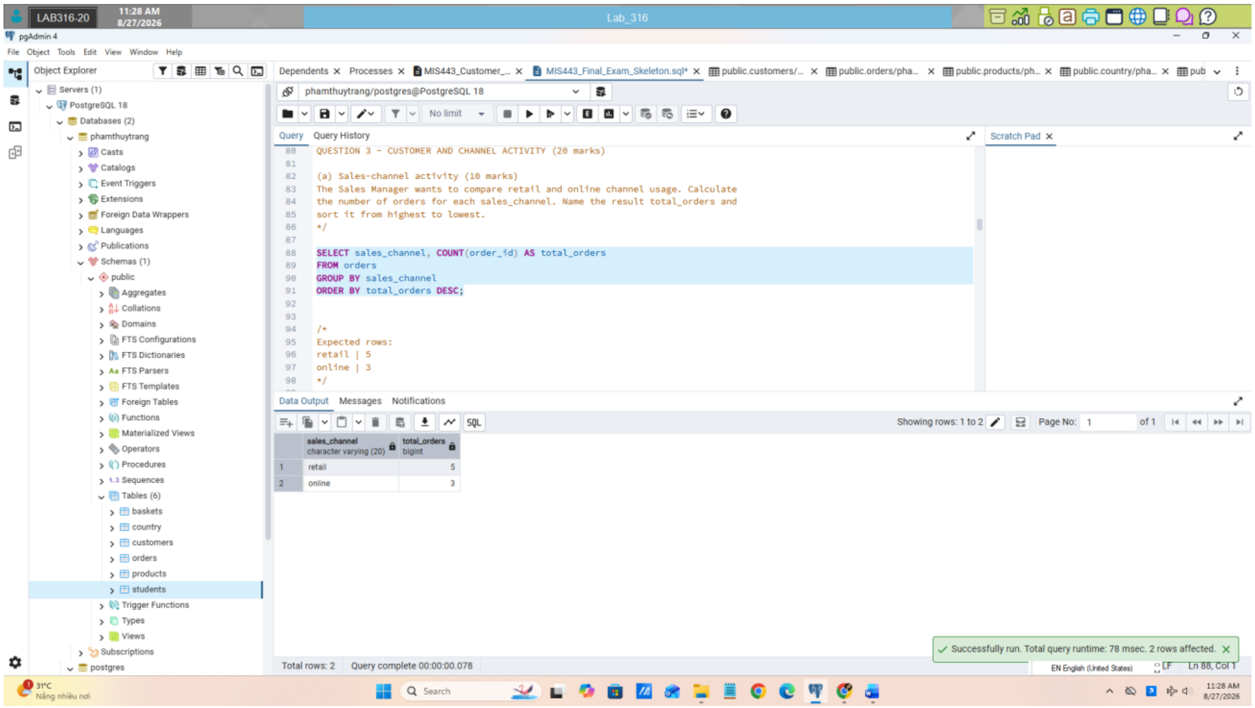
The Marketing Manager wants to understand customers who permit marketing emails. Calculate their average age, name the result average_age, and round it to two decimal places.




### Question 3 – Customer and Channel Activity (20 marks)

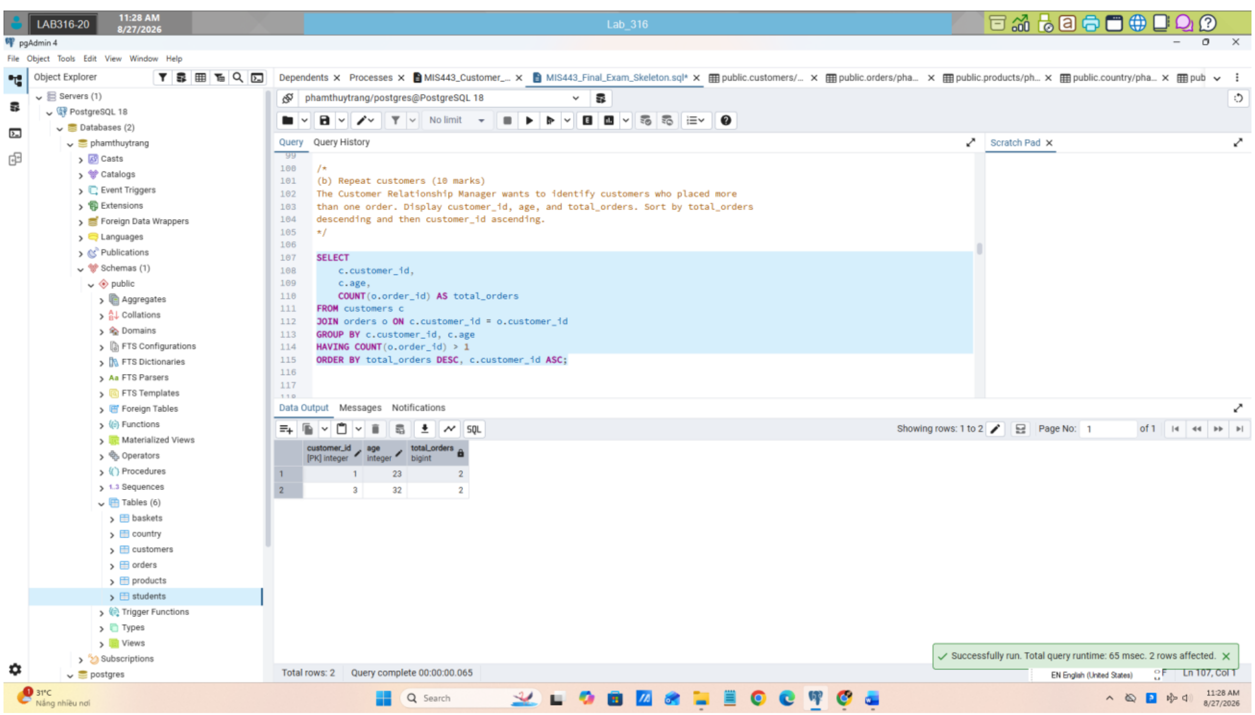
#### *(a) Sales-channel activity – 10 marks*

The Sales Manager wants to compare retail and online channel usage. Calculate the number of orders for each sales_channel. Name the result total_orders and sort it from highest to lowest.




## *(b) Repeat customers – 10 marks*

The Customer Relationship Manager wants to identify customers who placed more than one order. Display customer_id, age, and total_orders. Sort by total_orders descending and then customer_id ascending.



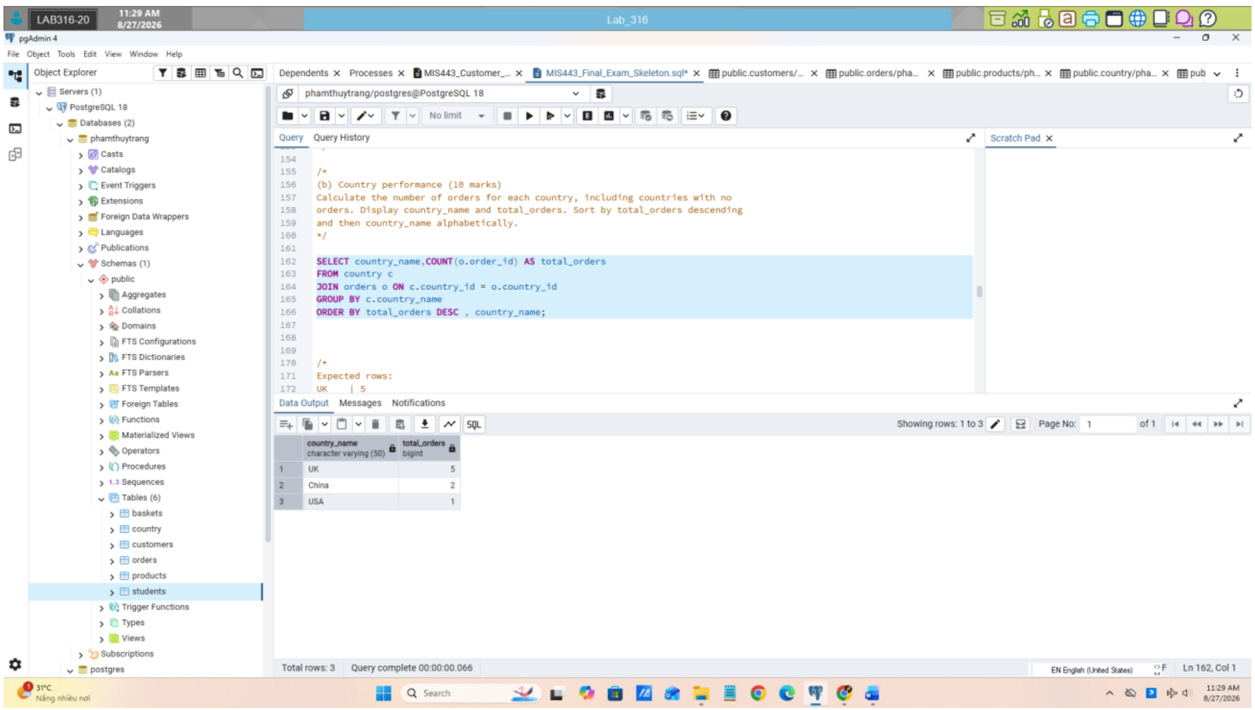
## Question 4 – Order and Country Analysis (20 marks)

### *(a) Order-monitoring report – 10 marks*

Create an order report showing order_id, date_shop, sales_channel, customer age as customer_age, and country_name. Sort by the newest order date and then by order_id.

### *(b) Country performance – 10 marks*

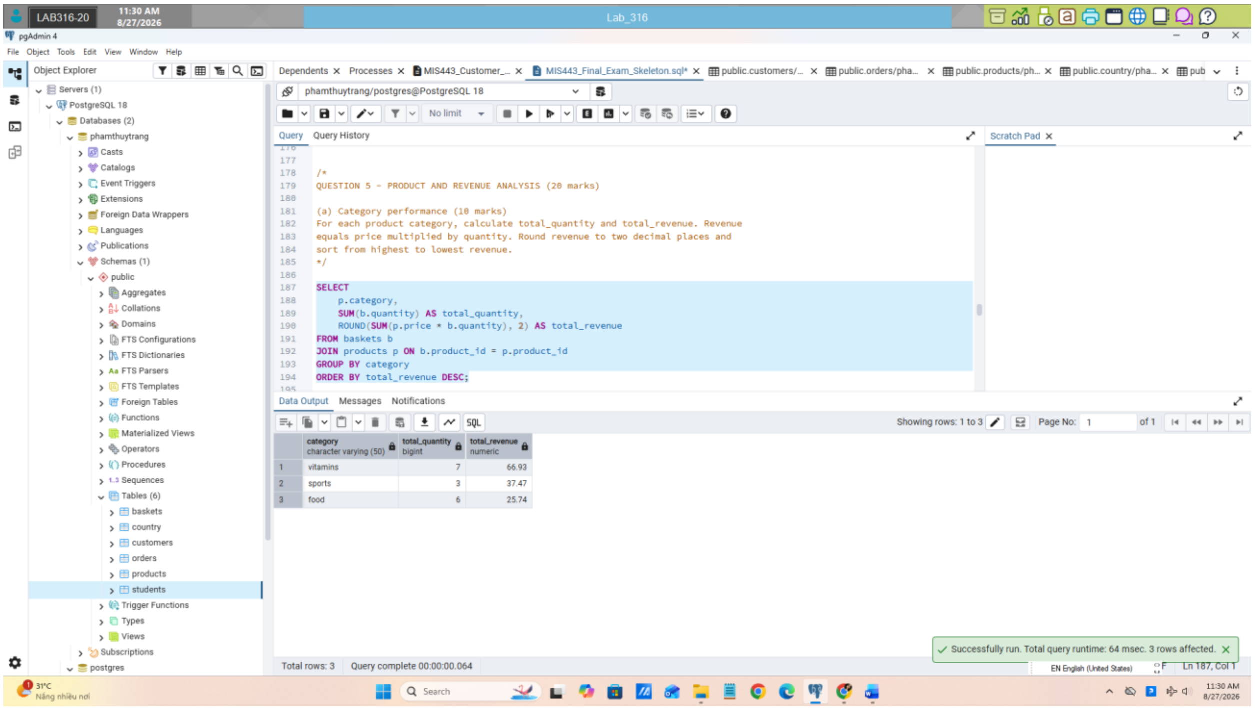
Calculate the number of orders for each country, including countries with no orders. Display country_name and total_orders. Sort by total_orders descending and then country_name alphabetically.




## Question 5 – Product and Revenue Analysis (20 marks)

### *(a) Category performance – 10 marks*

For each product category, calculate total_quantity and total_revenue. Revenue equals price multiplied by quantity. Round revenue to two decimal places and sort from highest to lowest revenue.




### *(b) High-value orders – 10 marks*

Calculate the total value of each order and return only orders worth more than 20.00. Display order_id and total_order_value, rounded to two decimal places. Sort from highest to lowest value.

### Question 6 – Advanced Business Analysis (20 marks)

### *(a) Customer purchase recency – 10 marks*

For every customer who permits marketing emails, display customer_id, age, and their latest order date as latest_order_date. Include eligible customers with no orders and sort by customer_id.

### *(b) Customer activity ranking – 5 marks*

Rank all customers by their number of orders. Customers with equal totals must receive the same rank without ranking gaps. Include customers with no orders.

Display customer_id, age, total_orders, and activity_rank. Do not use a CTE.

### *(c) Channel performance using a CTE – 5 marks*

Use a CTE to calculate each order's total value. Then summarise performance by sales_channel.

Display:

- sales_channel;
- total_orders;
- total_revenue;
- average_order_value.

Round monetary values to two decimal places and sort by total_revenue descending